

# Sebastián Gutiérrez Hernández

✉ shern3@gatech.edu • <https://sebasguthdz.github.io/>

## Professional Summary

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Computational mathematician specializing in developing computational techniques for Optimal Transport and PDE control using generative models and machine learning.

## Education

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**Georgia Institute of Technology** Aug 2021–Dec 2026 GPA 3.97  
*PhD in Computational Sciences and Engineering* Atlanta, GA  
Expected completion: December 2026  
Research focus: Flows in density space, generative algorithms, optimal control, and operator learning for PDEs

**Centro de Investigación en Matemáticas** Aug 2019–Jul 2021  
*M.Sc. in Applied Mathematics* Guanajuato, Mexico  
Thesis: Local Discontinuous Galerkin Methods for Deterministic and Stochastic Parabolic Conservation Laws

**National Polytechnic Institute of Mexico** Aug 2013–Jun 2018  
*B.Sc. in Physics and Mathematics* Mexico City, Mexico  
Sotero Prieto Prize honorable mention (2019) for B.Sc. dissertation

## Professional Experience

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**Mitsubishi Electric Research Laboratories** Summer 2024  
*Research Intern* Cambridge, MA  
Supervisor: Dr. Mouhacine Benosman  
○ Uncertainty quantification for physics-informed neural networks and reduced-order models. Delivered working proof-of-concept following mid-internship project pivot

**Georgia Institute of Technology** Spring 2022–Present  
*Graduate Research Assistant* Atlanta, GA  
Advisor: Dr. Haomin Zhou  
○ I research the use of a parametric optimization framework for density transport problems using statistical manifolds and generative models. Our first published work achieves up to 7% lower action, up to 10× improved boundary accuracy, and 40–80% faster runtimes compared to state-of-the-art baselines

**Georgia Institute of Technology** Fall 2023–Present  
*Graduate Research Assistant* Atlanta, GA  
Advisor: Dr. Peng Chen  
○ Developing machine learning surrogates (DeepONets and MIONETs) for uncertainty-dependent PDE-constrained optimization problems. Preliminary results demonstrate 43× speedup in solving constrained optimization problems compared to traditional methods

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| <b>Georgia Institute of Technology</b><br><i>Lecturer</i><br>Department: Mathematics<br>○ Integral Calculus                                                                                                                                                                               | Summer 2025<br>Atlanta, GA                  |
| <b>Georgia Institute of Technology</b><br><i>Graduate Research Assistant</i><br>Advisor: Dr. Rachel Kuske<br>○ Conducted spatial-temporal data analysis of competing species microbial models from 2D PDEs and discrete individual-based models                                           | Spring 2022–Fall 2022<br>Atlanta, GA        |
| <b>Centro de Investigación en Matemáticas</b><br><i>Graduate Research Assistant</i><br>Advisor: Dr. Silvia Jerez<br>○ Developed numerical solutions for stochastic PDEs in 1D, specializing in convection-dominated transport equations using Galerkin and Discontinuous Galerkin methods | Fall 2020–Spring 2021<br>Guanajuato, Mexico |
| <b>Technomadic</b><br><i>Data Science Intern</i><br>Supervisor: Anibal Bustillo<br>○ Developed ML models for time series prediction in agricultural applications                                                                                                                          | Aug–Sep 2018<br>Remote                      |

## Publications

1. **S. Gutierrez**, P. Chen, and H. Zhou. PDPO: Parametric Density Path Optimization. *NeurIPS* 2025.
2. **S. Gutierrez**. Local Discontinuous Galerkin Methods for Deterministic and Stochastic Parabolic Conservation Laws. Master’s thesis, CIMAT, 2021.

## Teaching & Leadership

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| Teaching Experience.....                                                                                                                                                                                                                                                             |
| <b>Fall 2021–Present: Teaching Assistant</b> , Georgia Institute of Technology<br>Multivariate Calculus (Fall 2021, Spring 2022, Fall 2022), Linear Algebra (Summer 2023), Ordinary Differential Equations (Spring 2023, Fall 2023, Fall 2025)                                       |
| <b>Fall 2024, Fall 2025: Lecture Assistant</b> , Georgia Institute of Technology<br>Partial Differential Equations (Fall 2024), Stochastic Process 1 (Fall 2024), Computational Methods for Simulation and Machine Learning (Fall 2025)                                              |
| Leadership Experience.....                                                                                                                                                                                                                                                           |
| <b>Fall 2024–Present: Student SIAM Chapter Vice President</b> , Georgia Institute of Technology<br><b>Fall 2023–Present: Mentor, Directed Reading Program</b> , Georgia Institute of Technology<br>Mentees: Arnav Hiray (Fall 2023–Spring 2024), James Iorio (Fall 2024–Spring 2025) |

## Technical Skills

**Machine Learning:** PyTorch, JAX, flow matching, DeepONets, MIONETs, physics-informed neural networks

**Scientific Computing:** NumPy, SciPy, FEniCS, finite element methods  
**Development Tools:** Python, C++, Linux, LaTeX, Git  
**Languages:** Spanish (native), English (fluent, TOEFL: 104/120), French (intermediate)

**Conference Presentations**

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Talks.....

**2024:** “Control-DeepONet: Deep operator networks for PDE-constrained optimal control under uncertainty,” SIAM UQ 2024  
**2021:** “Local Discontinuous Galerkin Methods for Deterministic and Stochastic Parabolic Conservation Laws,” CIMAT Applied Mathematics Seminar

Posters.....

**2024:** “Control Problems in Wasserstein Space via Conditional Flow Matching,” Workshop on Optimal Transport, Humboldt University Berlin

**Awards and Honors**

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**2024:** Outstanding Lecture Assistant, Georgia Institute of Technology  
**2023:** David L. Brown Fellowship, Georgia Institute of Technology  
**2019:** Honorable mention for bachelor’s thesis, Mexican Mathematical Society’s Sotero Prieto Prize